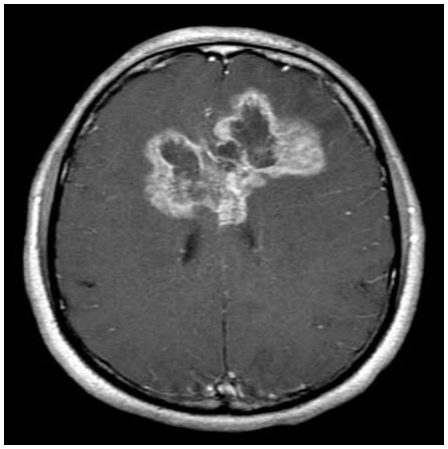
# NEURO-AI DIAGNOSTIC CLINICAL REPORT

**METAHEURISTIC OPTIMIZED AI**
Confidential Medical Document

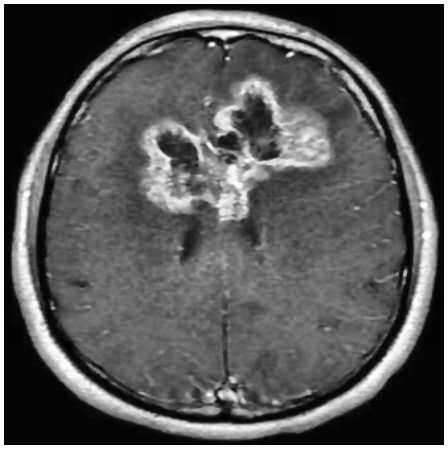
| | | |
|---|---|---|
| **Patient ID:** PAT-TEST01 | **Patient Name:** Alexander Wright | **Scan Date:** 2026-08-21 15:23:27 |
| **Modality:** Brain MRI (T1-CE / Axial) | **AI Engine:** Metaheuristic Multi-CNN | **Architecture:** ENSEMBLE |

**NORMAL SCAN — NO TUMOR DETECTED (Confidence: 26.7%)**

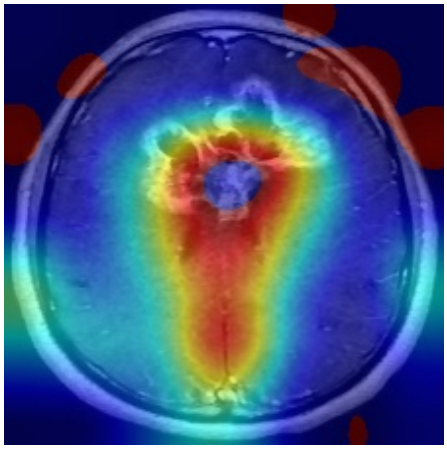
## MULTI-STAGE MRI VISUALIZATION & EXPLAINABILITY (XAI)



**1. Raw MRI Acquisition**
Unprocessed input scan



**2. Skull-Stripped & CLAHE**
Contour cropped & denoised



**3. Grad-CAM++ Saliency**
AI Attention Tumor Map

## DIAGNOSTIC PROBABILITY DISTRIBUTION & TUMOR LOCALIZATION

| Pathology / Class | Probability | Clinical Risk |
|---|---|---|
| Glioma | **23.2%** | High |
| Meningioma | **26.0%** | Moderate |
| Pituitary Tumor | **24.1%** | Moderate |
| Healthy / No Tumor | **26.7%** | None |

| Quantitative Lesion Metric | Observation Value |
|---|---|
| Estimated Tumor Area Burden | **16.77%** of brain tissue |
| Anatomical Localization | **No Pathological Mass (Intact)** |
| Mass Effect Index | **None** |
| Pathological Description | No pathological mass effect, abnormal contrast enhancement, or structural lesion detected. Symmetrical cerebral hemispheres and normal ventricular size. |

**Attending Radiologist / Clinician Sign-Off:**

Signature & Stamp

**AI Assistance Disclaimer:**
*This report is generated by a deep learning metaheuristic framework as a clinical decision support tool. Final clinical diagnosis must be verified by a board-certified radiologist.*